When a Learned σ-Profile Is Not One: Diagnosing the Intermediate of a Physics-Embedded Solubility Model

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Abstract

Physics-informed predictors combine learned physical quantities through a fixed thermodynamic relation, which allows a prediction to be audited and a better value for one quantity to be supplied without retraining. We ask whether the values such a network places in the slot of a physical quantity are that quantity. The system is a graph network supplying the σ-profile consumed by COSMO-SAC to predict the solubility of solids, and every measurement is read against a control. The observations determine two of the fifty directions in which a profile can move. The network places screening-charge density where the reference tabulation carries none, concentrating in the windows that activate the model's hydrogen-bonding term, and the output layer is not the cause; that departure is the most visible and the least consequential, since removing it leaves the thermodynamics unchanged. What the thermodynamics responds to is the magnitude of the profile where it is populated. The intermediate also does not decompose: replacing part of it with reference values is frequently worse than replacing none. Supplying the whole reference profile improves the activity coefficients the profile exists to produce and degrades the solubility predicted from them.

1 Introduction

How much of a solid dissolves in a given liquid decides whether a drug is formulated as a tablet or an injection, which solvent a crystallisation runs in, and how a product is purified. The value is measured one solute, one solvent and one temperature at a time. Most combinations of industrial interest have therefore never been measured, and a laboratory that needs one is choosing between running an experiment and trusting a prediction. The same is true across process chemistry, where a solvent is selected for a separation long before anyone measures the relevant phase behaviour.

Two kinds of prediction are on offer. A direct model regresses solubility from molecular descriptors or from a representation learned off the molecular graph, and returns a number. A physics-informed model predicts the quantities a thermodynamic argument names, then combines them through a fixed relation taken from that argument. The second kind has become popular for reasons that have little to do with accuracy, on which it usually trails.[1,2] Its internal quantities are things chemistry already knows how to measure and tabulate, so a prediction can be taken apart and checked piece by piece. Each piece can be trained against reference data that a direct regressor has no way to use. And the fixed relation supplies, for free,

structure that a purely statistical model would have to learn from data it does not have, which is the situation whenever the molecule in question resembles nothing in the training set.

That design rests on an assumption so natural it is rarely written down. If the quantity inside the model is a real physical quantity, then a better value for it can be looked up and handed to the model, and the prediction should improve. The assumption is what makes the internal quantity worth having, because it is the difference between a model that can be audited and repaired piece by piece and an elaborate curve fit with suggestive variable names. It also carries direct practical weight: it is the reason a laboratory would pay for a quantum-chemical calculation on a new compound, or curate a table of measured melting properties, rather than spend the same effort on more solubility measurements.

The quantity this work examines is the σ-profile. In the conductor-like screening model for real solvents and in its segment-activity reformulation COSMO-SAC,[3] a molecule enters the thermodynamics only through the distribution of screening-charge density over its cavity surface, that is, through how much of that surface carries charge density of each value. Everything else about the compound is discarded. The distribution comes from a quantum-chemical calculation with a continuum solvation treatment and is tabulated once per molecule.[4] It is an unusually attractive thing to learn: it is a fixed-length vector, it is tabulated for thousands of compounds, and supplying it from structure removes the only expensive step in the pipeline.

Group-contribution schemes supplied it first,[5,6] and learned predictors followed, taking the profile from a line notation or a molecular graph.[7–9] More recent work embeds the learning deeper, replacing both the profile and the segment-activity calculation with networks trained inside the thermodynamic structure, and reports accuracy beyond the model it is built from.[10] Others keep the thermodynamics and dispense with the named intermediate altogether,[11] or learn a thermodynamic potential directly from phase behaviour. Across this literature the internal quantity is described as physically meaningful on the grounds that it appears in the thermodynamic formulation. Validation, where it is reported for the intermediate at all, compares averaged predicted profiles against averaged reference profiles, or rests on the accuracy of the final prediction.

What has not been asked is whether the values a network places in that slot are the physical quantity. The question is not rhetorical. A solubility measurement constrains the combination of the intermediate with the fixed relation, not the intermediate on its own, so different internal values can reproduce the same measurements equally well; the directions the data leave undetermined are where a flexible network is free to move. Machine learning has met this before under a different name. In models that predict human-meaningful concepts and then predict the target from the concepts alone, the concepts leak. They carry information their labels do not name, and interventions on them behave in ways the names do not predict.[12–15] The thermodynamic case differs in one respect that makes it tractable. A σ-profile is not a human label but a computable quantity, so the leak can be measured against the quantity itself.

This work measures it. We examine a pipeline that predicts the solubility of a solid through COSMO-SAC, with a graph network supplying the σ-profile, trained both against tabulated reference profiles and through the closure. Solubility data do not determine the quantities the model composes. The network places screening-charge density where the reference tabulation carries none, and does so preferentially in the windows the hydrogen-bonding term reads; that departure turns out to be the most visible and the least consequential of those reported here. The output layer is not the cause: the same network reaches far smaller values elsewhere in the same profile, and an untrained network of the same architecture is worse by more than an order of magnitude. A molecule's profile depends on whether it enters as solute or as solvent, so the predicted activity of a molecule in itself departs from unity. What misplaced density costs the prediction varies

by more than an order of magnitude along the charge-density axis and steps at the hydrogen-bonding threshold. The learned profile does not decompose: replacing part of it with the reference values is frequently worse than replacing none. And the error that reaches the thermodynamics lives in the magnitude of the profile where it is populated, not in the spurious density where it should be empty.

This article is organised as follows. Section 2 describes the thermodynamic relation being closed, the network that supplies its input, the reference profiles and the data. Section 3 reports the measurements, each against the control that would refute it. Section 4 draws out what they imply for the design of models of this kind, and Section 5 states the limits of the evidence. Section 6 summarises.

2 The system under study

2.1 The relation being closed

The solubility of a crystalline solid in a liquid follows from a balance between two quantities. Breaking the crystal costs an amount set by the melting temperature T_m and the enthalpy of fusion ΔH_{fus}; mixing the freed molecules into the solvent is governed by the activity coefficient γ_2 of the solute in that solvent. For an ideal solution the two combine into the mole fraction at saturation,

$$\begin{aligned} \ln x_2 &= -\,\Phi(T) \;-\; \ln \gamma_2, \\ \Phi(T) &= \frac{\Delta H_{\mathrm{fus}}}{R}\left(\frac{1}{T}-\frac{1}{T_m}\right), \end{aligned} \tag{1}$$

with R the gas constant. The first term is the crystal contribution and carries no information about the solvent; the second is where the chemistry of the mixture enters. Because γ_2 itself depends on x_2, equation (1) is solved rather than evaluated.

COSMO-SAC[3] supplies γ_2. It represents each molecule by its σ-profile: the area of the molecular cavity surface carrying screening-charge density σ, discretised onto a fixed grid of 51 bins spanning roughly $\pm 0.025\, e\,\text{Å}^{-2}$. A molecule's identity enters the model through this vector and through the total cavity area; nothing else about the compound is used. Segments exchange with a common pool, and a self-consistent calculation returns a segment activity coefficient for each bin, from which the molecular activity coefficient follows. The exchange energy contains a term that switches on only for segments beyond a threshold σ_{hb} in magnitude, which is how the model represents hydrogen bonding; a segment qualifies by its charge density alone, with no reference to which atom carries it. That threshold divides the charge-density axis into a core and two outer windows, and the division recurs throughout what follows.

2.2 The network that supplies the profile

A graph neural network reads the solute and the solvent and produces, for each, a σ-profile and a cavity area (Figure 1); these are passed to the COSMO-SAC calculation and to the solver for equation (1). A second head predicts the crystal properties T_m and ΔH_{fus} that enter $\Phi(T)$. The profile head is a feed-forward network on the per-molecule representation, ending in a normalised, non-negative vector of 51 bins. The same head serves both roles, but reads a role-dependent representation, so a compound receives one profile as solute and another as solvent.

The head is trained in two ways at once, and this is the property that makes the pipeline suitable for the present question. It is supervised directly against tabulated reference profiles, as a predictor of those profiles would be. It is also trained through the thermodynamic relation, so the gradient of the solubility loss reaches it. The nearest prior model instead supervises its profile predictor externally and then freezes it, so that no gradient from the phase-equilibrium data reaches the intermediate.[10] The pipeline studied here therefore occupies the position where both influences act, which is where their relative effect can be read.

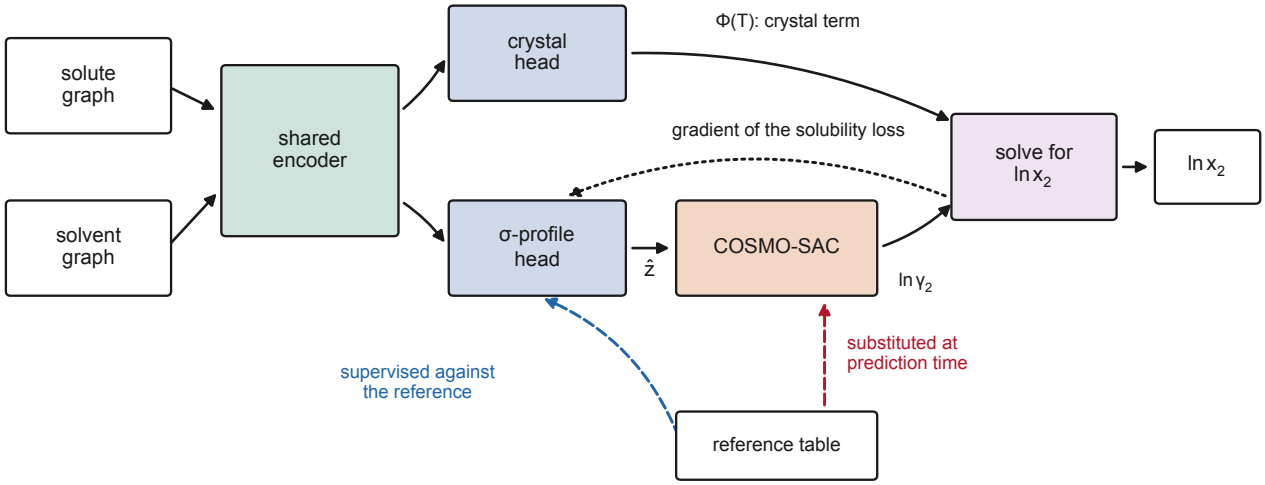


Figure 1: The pipeline. A shared encoder reads the solute and solvent graphs; one head supplies the crystal term of equation (1) and another the σ-profile $\hat{z}$ consumed by COSMO-SAC. The profile head is supervised against the reference table and also receives the gradient of the solubility loss. At prediction time the reference profile can replace the learned one.

2.3 Reference profiles, data and arms

Reference σ-profiles are taken from the VT-2005 tabulation,[4] computed by quantum chemistry with a continuum solvation treatment. Throughout, "reference" means a value from that table, and departures are measured against it. How far the tabulation itself depends on the underlying electronic structure method is a separate question[16] and is not at issue here, since every comparison in this work uses one table.

Solubility data are drawn from a curated compilation of measured solid solubilities,[17] split so that no scaffold[18] appears in both training and test sets, which is the harder of the two standard settings and the one relevant to new compounds. Infinite-dilution activity coefficients used to score the closure directly come from an independent compilation. All results below are reported over five training seeds; the arms differ only in the seed, so the spread across them is the natural scale against which any effect must be read. Architectural details, hyperparameters, the construction of the splits and the auxiliary supervision streams are given in the Supporting Information.

2.4 How the measurements are read

Each measurement in Section 3 is a contrast, and each is reported against a control chosen so that it would fire if the instrument responded to something other than the intended effect. Where a quantity is compared with a reference table, the control is the same quantity computed on profiles that carry the same total area but no chemistry. Where part of a profile is replaced, the control is a part of the same size chosen at random. Where an effect is attributed to training, the control is an untrained network of the same architecture. Each control, and the outcome that would refute the reading taken from it, is fixed with the measurement rather than after it, and the Supporting Information lists them; one measurement, the reference-state test, additionally carries a rule written down before it was computed, and that document is reproduced there. Table 1 collects every test with the control beside it, because a result of this kind is not interpretable without one.

Table 1: The diagnostic battery. Each test is read against the null named beside it; a result without its null is not a result. Numbers are medians over five training seeds unless a range is given.

Question	Instrument, and the null it is read against	Result	Reading
Does the observable determine the σ-profile?	spectrum of the closure's Jacobian over the 51 bins, 25 solvents $\times$ 5 seeds, against the same spectrum on random profiles of equal area	2 of 50 profile directions carry 99% of the information (range 1–5); random profiles give 2	No, and the deficiency is structural rather than a property of the trained head.
Does it determine the crystal term?	Fisher information of the four-parameter fit; Cramér–Rao bound on the enthalpy of fusion, against the same bound with external labels supplied	condition number 2.3×10^5; bound $\pm 20\,\mathrm{kJ\,mol^{-1}}$, falling to ± 1.7 with the labels	No. The crystal enthalpy is collinear with the activity's temperature slope.
Does the head use the freedom?	residual of learned minus reference profile by charge-density window, 71 molecules, against each window's share of surface area	donor window 4.0$\times$, acceptor 2.0$\times$, nonpolar core 0.88$\times$; 58% of reference cells are exactly zero	Yes, and into the windows the hydrogen-bonding term reads.
Is the output layer responsible?	mass in bins where the reference is exactly zero, against the smallest bin the same head reaches anywhere and against an untrained head	head reaches 10^{-11}; 34–51% of those bins already below 10^{-5}; pedestal 0.02–0.03 against 0.57 untrained	No. The pedestal is learned, and it is the same in both roles.
Is the reference state preserved?	activity of a molecule in itself, against the one-profile control that must return zero identically	control 0 exactly; median $\lvert\ln\gamma\rvert$ 0.012–0.044, largest 0.80	No. A molecule's two role profiles do not coincide.
What does misplaced mass cost, and where?	mean $\lvert\Delta\ln\gamma\rvert$ from moving a fixed mass into each bin, against random profiles of the same area	edge-to-core ratio 21$\times$ and 15$\times$ on the two row sets, against 1.6$\times$ for random profiles	Strongly non-uniform, with a step at the bonding threshold.
Does the intermediate decompose?	reference shape substituted into part of the grid, against a randomly chosen part of the same size	the whole shape helps on 5 of 5 seeds; a random third is worse than the fully learned profile in 20% of 20$\times$5 draws	No. The parts do not add, so per-bin attribution is undefined.
Where does the error that reaches the closure live?	reference shape in the bins where it is nonzero, against exact zeros in the bins where it is zero	+89% of the gap recovered against -9%	In the magnitude of the occupied bins, not in the pedestal.
What does supplying the reference profile do?	the same substitution scored on both quantities, against the 37 rows in which no profile is replaced because neither component is tabulated	activity coefficients $1.75 \pm 0.06 \to 0.76$; solubility worsens by 0.43 ± 0.04 over 5274 rows; placebo $< 4 \times 10^{-6}$	It improves the quantity the profile produces and degrades the quantity the pipeline predicts.

3 Results

Table 1 lists every measurement reported below together with the control it is read against. The order follows the question rather than the chronology: first whether the data determine the internal quantity at all, then what the network does with the freedom that remains, then what that costs the thermodynamics.

3.1 The observable determines two of fifty directions of the profile

A σ-profile is a point in a 51-dimensional space, and normalisation fixes one direction, so a profile can move in 50. The question is how many of those the closure exposes. For each solvent we collected the rows in which it appears, differentiated the predicted activity coefficient with respect to its profile, and took the spectrum of the resulting information matrix. Over 25 solvents and five seeds, the number of directions carrying 99% of the information is 2, with a range of 1 to 5; the median condition number is 1.4×10^{33}. The remaining directions are not merely poorly determined. They are flat to numerical precision, so a profile can move along them without changing any prediction the closure makes.

The same calculation on random profiles of equal area returns the same count. The deficiency is therefore structural, a property of the exchange kernel and of the diversity of partners available, rather than something the trained head produced. That is the stronger of the two readings: it holds wherever the profile is linearised, not only at the point this network happens to occupy.

The crystal term is underdetermined in the same way and for a reason that is easier to picture. The enthalpy of fusion and the temperature slope of the activity coefficient both enter equation (1) through $1/T$, so a fit can trade one against the other. For the four-parameter fit the condition number of the information matrix is 2.3×10^{5}, and no estimator can determine the enthalpy of fusion from a solubility series to better than about $\pm 20\ \mathrm{kJ\,mol^{-1}}$, which is wider than the range the quantity takes across these compounds. Supplying the melting temperature and the enthalpy of fusion as independent labels reduces that bound to $\pm 1.7\ \mathrm{kJ\,mol^{-1}}$.

What the network places in the undetermined directions is not decided by the data. The rest of this section measures what it places there.

3.2 The network places charge density where the reference carries none

The reference tabulation is sparse. A molecule with no hydrogen-bond donor is tabulated with no cavity surface carrying donor-side charge density, and 58% of all reference cells across the compounds examined here are exactly zero, a median of 29 bins of 51 per molecule.

The learned profiles are not sparse anywhere. Figure 2 shows the donor-window area for sixteen common solvents. Twelve of the sixteen are exactly empty in that window in the reference; all sixteen carry area there in the learned profile, between 0.1% and 5.2% of the molecule's surface. The hydrogen-bonding term reads that area regardless of how small it is, because a segment qualifies for the term by its charge density alone.

Where the misfit lands is not uniform across the charge-density axis. Take the residual between the learned and the reference profile over the 71 molecules that appear in a labelled test row, and express it relative to the share of surface area each window holds. The donor window is then over-represented by a factor of 4.0 and the acceptor window by 2.0, while the nonpolar core is under-represented at 0.88. The profiles a molecule receives as solvent give 3.8, 2.0 and 0.89 for the same three windows, so the orientation of the displacement does not depend on which role the molecule plays. The network's departure from the tabulation concentrates in the regions that switch on the model's hydrogen-bonding term and thins in the region that does not.

3.3 The output layer is not the cause

An obvious explanation is architectural. The head ends in a normalisation that cannot return an exact zero, so a profile that should be empty in a bin will instead be slightly populated there, and the defect would be an artefact of the parameterisation. The explanation is testable, and it fails.

Figure 3a shows, for every bin in which the reference is exactly zero, how much mass the trained head places there. The distribution reaches 1.0×10^{-11}, and between 34% and 51% of those bins already lie below 10^{-5}. A layer that can reach 10^{-11} is not what holds the remaining bins at 10^{-5}. Figure 3b makes the same point from the other direction. An untrained network of the same architecture places a median of 0.571 of the profile's mass in the bins where the reference is empty; after training the figure is 0.022 ± 0.004 over five seeds and both roles. The residual population is something training arrives at, not something the parameterisation imposes.

The two roles agree. The pedestal is 0.024 ± 0.004 on the profiles a molecule receives as solute and 0.020 ± 0.003 on those it receives as solvent, a difference of 0.004 against a seed-to-seed spread of the same size. This measurement and the one in Section 3.2 therefore hold separately for each of the two profiles a molecule is given, and do not depend on the fact that it is given two.

The point extends beyond the present pipeline. A recent physics-embedded model uses a rectified output with the same normalisation and can therefore emit exact zeros.[10] The measurement above indicates that the difference between the two output layers is not what decides whether a learned profile resembles the tabulated one.

3.4 A molecule is not ideal in itself

Any activity coefficient model must return unity for a pure component, since a molecule surrounded by copies of itself is in its reference state by definition. In COSMO-SAC the property holds identically: the residual term is a difference between a segment's activity in the mixture and in the pure compound, and the two coincide when one profile is used on both sides. Supplying a single profile to both sides returns exactly zero here, which confirms that the implementation carries the property.

The pipeline does not. Because the head reads a role-dependent representation, a compound receives one profile as solute and a different one as solvent, so pairing a molecule with itself pairs two different profiles. The median departure from ideality is between 0.012 and 0.044 in $\ln \gamma$ across the five seeds, the ninety-fifth percentile between 0.09 and 0.34, and the largest single value is 0.80. The violation belongs to the learned intermediate rather than to the thermodynamic relation. It is also invisible in any evaluation that never pairs a compound with itself, which is every evaluation on solubility or on mixture data.

3.5 What misplaced density costs, and where

The preceding measurements establish that density sits where the reference says it should not. They do not say what that costs, and the cost is not uniform. Figure 4a reports, for each bin, the mean change in the predicted activity coefficient produced by moving a fixed amount of mass from the core into that bin. Moving mass within the core is almost free. Moving the same mass into the outer windows costs 21 times as much as a median over seeds, with a range of 21 to 30; computed over the rows used for scoring rather than for training the ratio is 15, with a range of 13 to 25. Random profiles of the same total area give 1.6 and 1.5, so the structure reflects the chemistry of the exchange rather than the geometry of the grid.

The cost does not grow smoothly with distance along the axis. It is flat across the core, rises steeply as the hydrogen-bonding threshold is crossed, and then saturates. Once mass is inside the bonding region, pushing it further out changes the prediction comparatively little. The consequence is shown in Figure 4b. Read-

ing the measured cost as a potential and taking its gradient gives the weight under which the usual transport distance between two profiles would price their difference correctly, and that weight peaks at the boundary rather than deep in the outer window. A treatment that weighted the far corner most heavily, which is what the geometry of the grid would suggest, would weight the wrong place.

3.6 The intermediate does not decompose

If the learned profile were a σ-profile carrying a per-bin error, then correcting part of it should help in proportion to the part corrected. It does not. Replacing the learned shape with the reference shape over the whole grid improves the predicted activity coefficients on all five seeds. Replacing it over a randomly chosen third of the bins makes the prediction worse than leaving the profile entirely alone in 20% of draws, taken over twenty random thirds at each of five seeds and ranging from 5% to 30% by seed. Recovery of the gap across those draws runs from -213% to $+152\%$.

A profile assembled from correct values in some bins and learned values in the rest is not intermediate between the two. The segment-activity calculation is a self-consistent solution over the whole distribution, so an internally inconsistent profile can sit far from both of the profiles it was built from. Two things follow. The first is methodological: the share of the error attributable to any subset of bins is not defined on this surface, so an analysis that partitions the grid and attributes error to the parts cannot be carried out, whatever the partition. The second concerns the premise under test. Partial correctness of the intermediate is not partially useful, and is frequently worse than consistent incorrectness.

3.7 The error that reaches the thermodynamics is in the occupied bins

One partition is well posed, because both of its parts are substituted in full: the bins where the reference is nonzero, and the bins where it is zero. Figure 5a reports both. Supplying the reference values in the bins the reference occupies recovers 89% of the gap between the learned and the reference profile, positive on all five seeds and ranging from 68% to 109%. Setting exact zeros in the bins the reference leaves empty recovers -9% on average, with a per-seed range from -36% to $+34\%$, and the result sits inside the spread of a control that zeroes the same number of randomly chosen cells.

The spurious density is therefore not what the thermodynamics responds to. The error that propagates through the closure lives in the magnitude of the profile where the profile is populated. The cost map of Section 3.5 does not localise this error either. Figure 5b shows that substituting the reference into the third of the grid the closure is most sensitive to recovers no more of the gap than substituting into a randomly chosen third, and that the sign varies across seeds. Sensitivity per unit of displaced mass and total contribution to the error are different quantities, and the core carries most of the mass.

The profile's shape and the molecule's cavity area, which COSMO-SAC consumes as a separate scalar, split the gap unevenly. Replacing the whole shape while keeping the learned area moves the mean error from 1.75 to 1.08, against 0.76 when the reference profile is supplied with its own area, so the shape accounts for 0.68 of the gap and the area for the rest. This is a decomposition of two full substitutions rather than an attribution over bins, and is not subject to the objection raised in Section 3.6.

3.8 Supplying the reference improves one quantity and degrades the other

The substitution the premise licenses can be scored on two quantities at once: the activity coefficient the profile exists to produce, and the solubility the pipeline is built to predict. Both are measured on the same checkpoints.

On the activity coefficients, replacing the learned profiles with the reference profiles moves the mean absolute deviation from 1.751 ± 0.063 to 0.765, an improvement on every one of the five seeds and more than fourteen times the seed-to-seed spread of the arm. On the solubility, the same replacement at prediction time moves the error in the opposite direction, by 0.431 ± 0.044 in mean absolute error over the 5274 test rows in which a reference profile was available for the solvent, with per-seed values of 0.456, 0.381 and 0.457.

The control is internal to the same comparison. No reference profile exists for either component in 37 test rows, so nothing is substituted in them, and across those rows the two arms differ by at most 4×10^{-6} in mean absolute error. The effect is therefore carried by the rows in which the substitution actually happened, and not by any difference in how the two arms were scored. The substitution is a solvent-side substitution throughout: of the 147 distinct solutes in the test rows only seven are tabulated, so no row exists in which the solute profile alone is replaced. What is reported is what the available reference data permit, and the solvent channel is where the comparison lives.

The two directions are the result this work was undertaken to obtain. A better value for the internal quantity makes the thermodynamic model it feeds more accurate and makes the pipeline less accurate.

4 Discussion

The measurements describe a quantity that occupies the place of a σ-profile without being one. It is sparse where the reference is dense and populated where the reference is empty. It depends on the role the molecule plays in the pair, so the model it feeds does not return the reference state. And it does not decompose, so the familiar move of correcting part of it and expecting part of the benefit is unavailable.

The sharpest statement of the situation is a contrast between two substitutions of the same values. Supplying the reference profile improves the quantity the profile exists to produce: the predicted activity coefficients move most of the way toward the measured ones, on every seed. The same substitution degrades the solubility the pipeline is built to predict. Both facts are measured on the same checkpoints and the same compounds. Taken together they say that the downstream parts of the model have adapted to the profile the network supplies rather than to the one the physical argument names. Handing the model a physically better value therefore moves it away from the operating point it was fitted at. This is the behaviour the concept-bottleneck literature reports for learned concepts in general,[12,13] arrived at here through a thermodynamic decoder and, because the concept has a computable ground truth, measured against the concept itself rather than inferred from how interventions behave.

It follows that the assumption set out in the Introduction is not a property of physics-informed models as such but a property of how the internal quantity is trained. The design choice that matters is whether the gradient of the target reaches the intermediate. Where it does not, as in a model whose profile predictor is supervised externally and then frozen,[10] the intermediate should remain what it is named after to within its own fit error, and substitution should be a sensible operation. That expectation is not tested here, since the model in question was not retrained. Where it does, the intermediate becomes a free internal coordinate that happens to be initialised and regularised toward a physical quantity, and the operations the physical reading licenses stop being safe. The present pipeline sits between the two, with both influences acting, and the measurements show which one wins.

For practice this suggests a different division

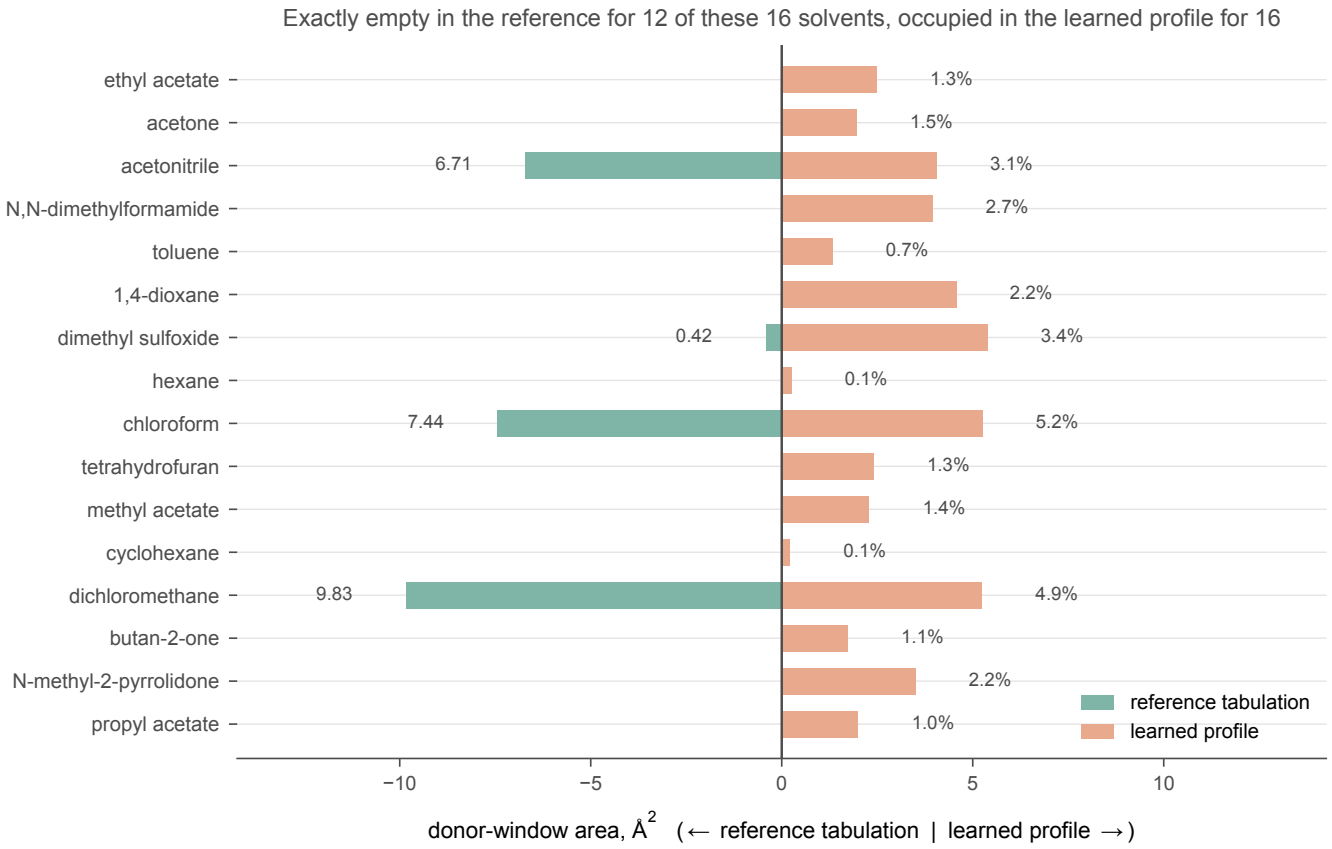


Figure 2: Donor-window area for sixteen solvents in the reference tabulation (left) and in the learned profile (right). Percentages give that window's share of the molecule's cavity surface in the learned profile.

of labour than the one the literature has settled into. If the internal quantity is to be read as chemistry, exchanged with tabulated values, or audited against reference data, it should be supervised against that reference and then held fixed. The capacity needed to fit the data belongs somewhere that makes no physical claim. If instead the intermediate is allowed to move under the target's gradient, the gain in accuracy is real but the physical reading has to be dropped, and with it the argument that recommends these models over direct regression in the first place. What is not available is to train the intermediate end to end and retain the interpretation, which is the position most of this literature currently occupies.

Three measurements in the battery are worth separating from the rest because each removes an explanation that would otherwise be reached for. The output layer is not responsible for the spurious density: the same head reaches values six orders of magnitude smaller elsewhere in the same profile. The cost map does not localise the error: the bins the closure is most sensitive to account for no more of the gap than a randomly chosen third. Sensitivity is measured per unit of displaced mass, and most of the mass sits in the insensitive core. And the spurious density in the empty bins, the most visible defect and the one easiest to describe, is not what the thermodynamics responds to; the error that propagates is in the magnitude of the profile where the profile is populated.

The finding is conditional on the fixed relation being an adequate description of the chemistry it is applied to. Where a relation is well specified for its domain, supplying it with more accurate inputs behaves as the physical reading predicts. A second system, in which a linear free-energy relation stands in for the closure and its adequacy varies by chemical stratum, separates in that way; it is reported in the Supporting

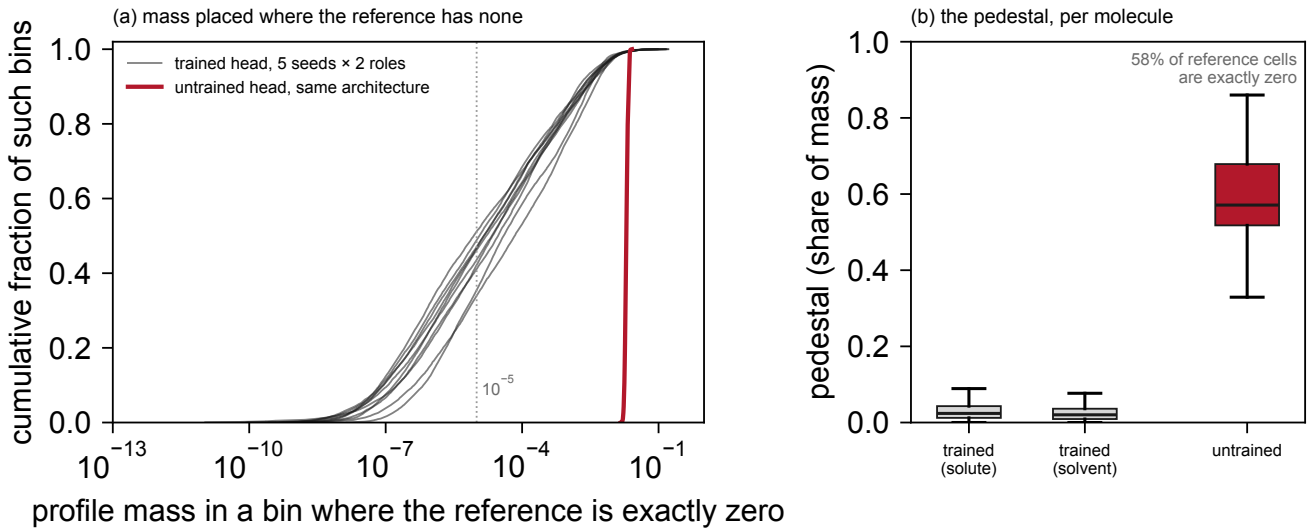


Figure 3: (a) Cumulative distribution of the profile mass placed in bins where the reference is exactly zero, for five seeds in both roles and for an untrained network of the same architecture. (b) The same mass summed over each molecule.

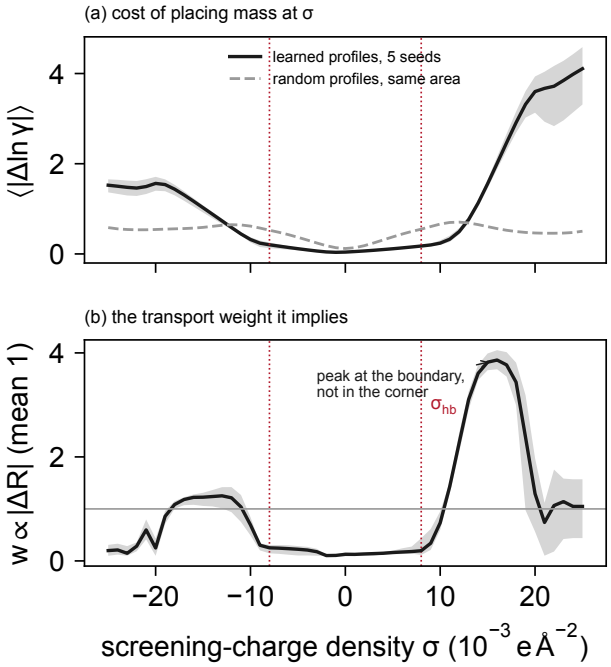


Figure 4: (a) Mean change in the predicted activity coefficient produced by moving a fixed mass from the core into each bin, for five seeds (band) and for random profiles of the same total area. (b) The gradient of that cost, normalised to mean one. Dotted lines mark $\pm\sigma_{\mathrm{hb}}$.

Information. The result here should therefore be read as a statement about an intermediate trained through an approximate closure, which is the common case, rather than as an argument against building physics into a model.

5 Limitations

The evidence comes from one pipeline, one closure and one reference tabulation. Every departure reported above is measured against the VT-2005 table. How far that table itself moves with the electronic structure and solvation treatment behind it is a separate question.[16] A different tabulation would shift the reference against which the learned profile is scored, but not the comparisons between arms, which all use one table.

A zero in that table is a tabulated zero. The σ-profile is built by binning a finite set of surface segments, so a bin can be empty because no segment of the molecule falls in it as well as because the molecule cannot carry charge density of that value. The distinction does not affect the comparisons reported here, all of which treat the tabulation as the reference. It does bound how strongly the finding may be stated: what is measured is a departure from the tabulated distribution, not a departure from a proven absence of density.

The profile head reads a role-dependent representation, so a molecule receives two profiles.

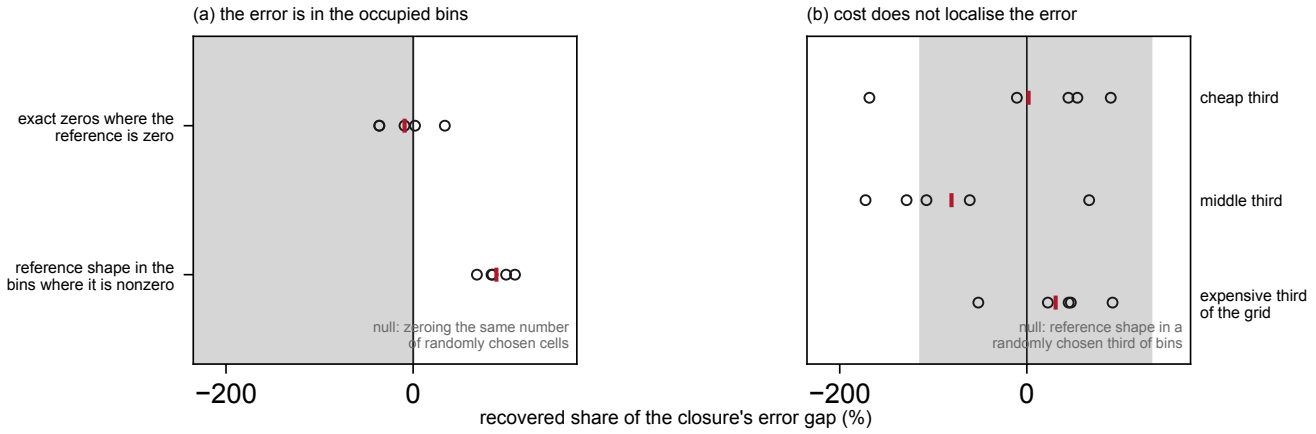


Figure 5: Share of the gap between the learned and the reference profile recovered in the predicted activity coefficients, one circle per seed and a bar at the mean. (a) Reference values supplied in the bins where the reference is nonzero, and exact zeros in the bins where it is zero. (b) Reference values supplied in thirds of the grid ordered by the cost of Figure 4. The shaded band is the central 90% of the control named in each panel.

That is a choice made in this pipeline, and a reader is entitled to ask whether the defects reported above are consequences of it. For two of them the question is settled by measurement. The window displacement of Section 3.2 and the residual population of Section 3.3 are the same on each of the two profiles taken separately, with the difference between roles no larger than the spread across seeds. The violation of the reference state in Section 3.4 is a direct consequence of the choice and would disappear under a role-independent head, at a cost in accuracy that is not measured here. The non-additivity of Section 3.6 is a property of the segment-activity calculation rather than of the head, since it concerns how the closure responds to an internally inconsistent profile. An arm trained with a role-independent head would settle the remaining cases directly, and has not been run.

Equation (1) omits the heat-capacity difference between the solid and the subcooled liquid, as is usual. Where the temperature lies far below the melting point that term is not negligible, and it adds a further source of indeterminacy to the crystal contribution beyond the collinearity reported in Section 3.1.

The activity coefficients used to score the closure are at infinite dilution, while the solubility the pipeline predicts is a finite mole fraction at saturation. The two are the same functional of the same profiles evaluated at different compositions. The comparison in Section 3.8 therefore reports the direction of the substitution's effect on each, not a single error carried between them.

This work does not report a more accurate solubility model. On the scaffold-disjoint split used throughout, the physics path trails a matched direct control sharing the same encoder in the mean over seeds, and both sit near a random-forest baseline. Both are well behind the leading direct models on extrapolation to new solutes,[1] and absolute errors are within reach of the disagreement between independent measurements of the same system. No published model was retrained here, so the comparison with prior work is a comparison of design choices rather than of performance.

Results are reported over five training seeds, and for some quantities that is not enough. The cost of the spurious density, expressed in the error of the predicted activity coefficients, varies across seeds by several times the spread of the arms themselves. Its sign is therefore not determined by the present evidence. What is determined is that it is not large and positive. The asymmetry of the cost map between the two sides of the charge-density axis depends on which set of molecule pairs it is computed over,

and is reported for both.

The closure is the 2002 formulation of COSMO-SAC with a single hydrogen-bonding term, run without the combinatorial contribution. That term needs molar volumes, which the source database carries but the extracted table used here does not, and which the network could supply with a second scalar head beside the one that already predicts cavity area. Its omission is a choice made when the table was extracted rather than a constraint. The term vanishes for a pair of a molecule with itself, so Section 3.4 is unaffected; it would change the magnitudes in Sections 3.5 and 3.7, where the size of the effect is the result. Later formulations split the hydrogen-bonding term by segment type and would change where on the charge-density axis the cost steps, though not the existence of a step. The segment calculation is iterated to a fixed count; the count used at training and at scoring is recorded with each run, because the two need not agree and the predicted activity coefficients depend on them.

Finally, the measurements concern substitution at prediction time on an intermediate that was trained through the closure. They do not establish what would happen in a model whose intermediate is supervised externally and frozen. That design is the one the results recommend, and testing it requires training such a model rather than measuring this one.

6 Conclusions

A network trained inside a fixed thermodynamic relation fills the relation's input slot with values that are not the physical quantity the slot names. The learned σ-profile places screening-charge density where the reference tabulation carries none, concentrating in the windows that trigger the model's hydrogen-bonding term. It differs depending on whether the molecule is treated as solute or solvent, so a molecule is not predicted to be ideal in itself. And it cannot be corrected in parts: replacing a fraction of it with reference values is often worse than replacing none. The output layer is not the cause, and the error that reaches the thermodynamics lies in the magnitude of the profile where it is populated rather than in the spurious density where it should be empty.

Supplying the reference profile improves the activity coefficients the profile exists to produce and degrades the solubility the pipeline predicts. The interpretation that makes physics-informed models attractive, that a better value for an internal quantity can be looked up and handed to the model, holds only where that quantity is supervised externally and shielded from the target's gradient. Where the intermediate is trained end to end, it is an internal coordinate with a physical name, and the operations the name licenses should be measured before they are used.

Author contributions

N. L. Polomoshnov: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft. A. V. Rudik: Writing – review & editing.

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Notes

The authors declare no competing financial interest.

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Parts of the analysis code and of the manuscript text were drafted with the assistance of a large language model, under author direction; all reported measurements were produced by code in the deposited repository and checked by the authors against the deposited artifacts.

Data and Software Availability

The analysis code, the deposited numerical results and the checks that bind the two are available at `https://github.com/doctawho42/learned-sigma-profile` under the MIT licence; the commit the reported numbers were produced from is recorded in the deposit. Model checkpoints and the processed split are deposited in the Zenodo archive `10.5281/zenodo.23907894`. The deposit covers the COSMO-SAC arms reported here; weights for the NRTL variants discussed in the Supporting Information were not retained. Solubility data are BigSolDB 2.0;[17] reference σ-profiles are the VT-2005 database[4] as redistributed with Ref.;[19] infinite-dilution activity coefficients are from ThermoML;[20] the acid-dissociation data used in the Supporting Information are the OPERA compilation.[21]

Supporting Information Available

Architecture and training details; construction of the splits and the auxiliary supervision streams; the control for each measurement, with the outcome that would refute it; what the observations leave undetermined, which is the premise the measurements rest on; and the second system based on a linear free-energy relation.

References

(1) Attia, L.; Burns, J. W.; Doyle, P. S.; Green, W. H. Data-driven organic solubility prediction at the limit of aleatoric uncertainty. *Nature Communications* **2025**, *16*, 7497, FASTSOLV model; earlier ChemRxiv 2024 preprint doi:10.26434/chemrxiv-2024-93qp3.

(2) Vermeire, F. H.; Chung, Y.; Green, W. H. Predicting Solubility Limits of Organic Solutes for a Wide Range of Solvents and Temperatures. *Journal of the American Chemical Society* **2022**, *144*, 10785–10797.

(3) Lin, S.-T.; Sandler, S. I. A Priori Phase Equilibrium Prediction from a Segment Contribution Solvation Model. *Industrial & Engineering Chemistry Research* **2002**, *41*, 899–913.

(4) Mullins, E.; Oldland, R.; Liu, Y. A.; Wang, S.; Sandler, S. I.; Chen, C.-C.; Zwolak, M.; Seavey, K. C. Sigma-Profile Database for Using COSMO-Based Thermodynamic Methods. *Industrial & Engineering Chemistry Research* **2006**, *45*, 4389–4415.

(5) Mu, T.; Rarey, J.; Gmehling, J. Group contribution prediction of surface charge density profiles for COSMO-RS(Ol). *AIChE Journal* **2007**, *53*, 3231–3240.

(6) Mu, T.; Rarey, J.; Gmehling, J. Group contribution prediction of surface charge density distribution of molecules for COSMO-SAC. *AIChE Journal* **2009**, *55*, 3298–3300.

(7) Kang, J.-L.; Chiu, C.-T.; Huang, J. S.; Wong, D. S.-H. A surrogate model of sigma profile and COSMOSAC activity coefficient predictions using transformer with SMILES input. *Digital Chemical Engineering* **2022**, *2*, 100016.

(8) Chen, G.; Song, Z.; Qi, Z. Transformer-convolutional neural network for surface charge density profile prediction: Enabling high-throughput solvent screening. *Chemical Engineering Science* **2021**, *246*, 117002.

(9) Zhang, J.; Wang, Q.; Shen, W. Message-passing neural network based multi-task deep-learning framework for COSMO-SAC based sigma-profile and VCOSMO prediction. *Chemical Engineering Science* **2022**, *254*, 117624.

(10) Yang, Y.; Lin, S.-T. Physics-Embedded Machine Learning Model for Phase Equilibrium Prediction in Multicomponent Systems. *Journal of Chemical Information and Modeling* **2025**, *65*, 10180–10193.

(11) Specht, T.; Nagda, M.; Fellenz, S.; Mandt, S.; Hasse, H.; Jirasek, F. HANNA: hard-constraint neural network for consistent activity coefficient prediction. *Chemical Science* **2024**, *15*, 19777–19786.

(12) Koh, P. W.; Nguyen, T.; Tang, Y. S.; Mussmann, S.; Pierson, E.; Kim, B.; Liang, P. Concept Bottleneck Models. Proceedings of the 37th International Conference on Machine Learning. 2020; pp 5338–5348.

(13) Margeloiu, A.; Ashman, M.; Bhatt, U.; Chen, Y.; Jamnik, M.; Weller, A. Do Concept Bottleneck Models Learn as Intended? *arXiv preprint arXiv:2105.04289* **2021**,

(14) Mahinpei, A.; Clark, J.; Lage, I.; Doshi-Velez, F.; Pan, W. Promises and Pitfalls of Black-Box Concept Learning Models. *arXiv preprint arXiv:2106.13314* **2021**,

(15) Havasi, M.; Parbhoo, S.; Doshi-Velez, F. Addressing Leakage in Concept Bottleneck Models. Advances in Neural Information Processing Systems (NeurIPS). 2022.

(16) Salih, F. Y. M.; Abranches, D. O.; Maginn, E. J.; Colón, Y. J. Open-source generation of sigma profiles: impact of quantum chemistry and solvation treatment on machine learning performance. *Digital Discovery* **2025**, *4*, 2711–2723.

(17) Krasnov, L.; Malikov, D.; Kiseleva, M.; Tatarin, S.; Sosnin, S.; Bezzubov, S. BigSolDB 2.0, dataset of solubility values for organic compounds in different solvents at various temperatures. *Scientific Data* **2025**, *12*, 1236.

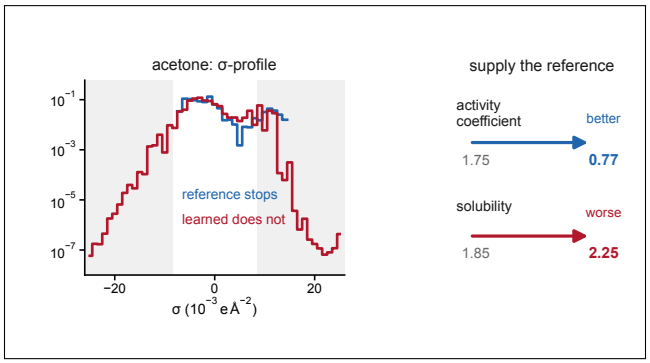
(18) Bemis, G. W.; Murcko, M. A. The Properties of Known Drugs. 1. Molecular Frameworks. *J. Med. Chem.* **1996**, *39*, 2887–2893.

(19) Bell, I. H.; Mickoleit, E.; Hsieh, C.-M.; Lin, S.-T.; Vrabec, J.; Breitkopf, C.; Jäger, A. A Benchmark Open-Source Implementation of COSMO-SAC. *Journal of Chemical Theory and Computation* **2020**, *16*, 2635–2646.

(20) Frenkel, M.; Chirico, R. D.; Diky, V. V.; Yan, X.; Dong, Q.; Muzny, C. ThermoML: An XML-Based Approach for Storage and Exchange of Experimental and Critically Evaluated Thermophysical and Thermochemical Property Data. *Journal of Chemical Information and Computer Sciences* **2003**, *43*, 1344–1359.

(21) Mansouri, K.; Cariello, N. F.; Korotcov, A.; Tkachenko, V.; Grulke, C. M.; Sprankle, C. S.; Allen, D.; Casey, W. M.; Kleinstreuer, N. C.; Williams, A. J. Open-source QSAR models for pKa prediction using multiple machine learning approaches. *Journal of Cheminformatics* **2019**, *11*, 60.

TOC Graphic


acetone: σ-profile
10^{-1}
10^{-3}
10^{-5}
10^{-7}
reference stops
learned does not
−20
0
20
σ (10^{-3} eÅ^{-2})
supply the reference
activity coefficient
better
1.75
0.77
solubility
worse
1.85
2.25